INCH-POUND

MIL-S-48685 (AR)
AMENDMENT 1
16 February 1990

MILITARY SPECIFICATION

SAFETY AND ARMING DEVICE, GUIDED MISSILE: M143E1
ELECTRICAL PARTS AND ASSEMBLIES

This amendment forms a part of Military Specification MIL-S-48685, dated 4 February 1987, and is approved for use within the US Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Defense.

PAGE 3

3.3.3.1, line 5: Delete "2.5" and substitute "1.5".

PAGE 5

3.4.1, Table II, Terminal E20 to E21, under requirements:

Delete "Less than 19 MEG: and substitute "Less than 500K".

3.4.1, Table II, Terminal E21 to E20, under requirements:

Delete "Greater than 20 MEG: and substitute "Less than 500K".

PAGE 9

J1-13, line 2, Under points requiring logic high: Delete "J17" and substitute "J1-17".

PAGE 10

J1-27, Under points requiring logic high: Delete "FR12" and substitute "FL2".

J1-33, Under points requiring logic high: Add "L20".

CASE Ground
(Hole #182), Under points requiring logic high: Delete "All low" and substitute "J1-50, L5".

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

PAGE 15

Table IX, g&h: Delete "2.5 (six places)" and substitute "4.Ø (six places)".

PAGE 16

3.7.1, line 7: Delete "2.5" and substitute "1.5".

3.7.2, line 6: Delete "2.5" and substitute "4.Ø".

3.7.4: Delete in its entirety and substitute

"3.7.4 Fire circuit function. The initiation of either the left or right fire circuits functioning is determined by the occurrence of the fire TM output changing from Ø ± .5V to 4.5 ± 1.ØV at (+) J1-2Ø to J1-15. The left fire circuit has completed functioning when the left fire circuit capacitor voltage (+) J1-38 to (-) J1-15 has discharge to less than 15. OVDC in no greater than 4ØØ milliseconds after the the occurrence of a fire TM output. The right fire circuit has completed functioning when the right fire capacitor voltage (+) J1-36 to (-) J1-15 has discharge to less than 15.OVDC in no greater than 4ØØ msec after the occurrence of a fire TM output."

PAGE 17

3.7.4.1.1: Delete in its entirety and substitute

"3.7.4.1.1 Loss of power after arming. The initiation of fire circuit functioning as defined by paragraph 3.7.4 shall occur in no less than 1Ø milliseconds and no greater than 5Ø milliseconds upon removal of power/destruct (+) J1-3Ø to (-) J1-15."

3.7.4.1.2: Delete in its entirety and substitute

"3.7.4.1.2 Fire upon arming after loss of power. The fire circuits shall function as described in paragraph 3.7.4 upon arming, after removal of power/ destruct (+) J1-3Ø to (-) J1-15."

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

PAGE 20

3.9.4: Delete in its entirety and substitute

"3.9.4 Semiconductor Burn-In."

Add:

"3.9.4.1 Diodes and Rectifiers. Devices shall be subjected to burn-in testing in accordance with the details herein and MIL-STD-750, Method 1038, Test Condition B and shall meet the final electrical requirements after exposure.

3.9.4.2 Transistors. Devices shall be subjected to burn-in testing accordance with details herein and MIL-STD-750, Method 1039, Test condition B and shall meet the final electrical requirements after exposure."

PAGE 22

4.3.1, Under drawings: Delete "12630705, 12630850 and 12630854" in its entirety.

4.3.1, Under drawings: Delete "12630851," and substitute "9311361".

4.3.3, line 1: Delete "componment" and substitute "component".

PAGE 37

4.4.2.10, Sheet 1 of 1: Delete in its entirety and replace with included sheet.

PAGE 38

4.4.2.10.1, Title: Add "Destructive Testing".

4.4.2.10.1, Drawing Number: Add "-1".

4.4.2.10.1, Notes: Add "1/ not required for 12630838-2".

PAGE 39

4.4.2.11, Sheet 2 of 2, next higher assembly: Delete "12630854" and substitute "12630849".

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-48685 (AR)
AMENDMENT 1

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 12630838/-1/-2/1/
4.4.2.10	Transistor, High Power, Silicon P.N.P 2N6193 or JANTX2N6193				NEXT HIGHER ASSEMBLY 12630829
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Burn-in		100%	3.9.4	Test
102	Collector cut-off current with emitter open		100%	3.2	Test
103	Emitter-cut-off current		100%	3.2	Test
104	Collector-to-emitter sustaining voltage with base open		100%	3.2	Test
105	Collector-to-emitter saturation voltage		100%	3.2	Test
106	Base-to-emitter voltage		100%	3.2	Test
107 1/	DC forward current transfer ratio		100%	3.2	Test
Minor					
201	Marking (after final electrical)		0.65%	3.10	Visual
202	Evidence of poor workmanship		1.0%	3.12	Visual

NOTES: 1/ For JANTX2N6193 MIL-STD-19500 applies and only category 107, 201, & 202 are required.

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

PAGE 40

4.4.2.12, Sheet 1 of 1, drawing number: Delete "12630851" and substitute "9311361".

PAGE 45

4.4.2.15, Sheet 1 of 2, paragraph: Delete "4.4.2.15" and substitute "4.4.2.15.1".

PAGE 46

4.4.2.15, Sheet 1 of 1, paragraph: Delete "4.4.2.15" and substitute "4.4.2.15.1".

PAGE 50

4.4.2.21, Sheet 1 of 1, next higher assembly: Delete "12630854" and substitute "12630849".

PAGE 51

4.4.2.22, Sheet 1 of 1: Delete in its entirety.

PAGE 55

4.4.2.26, Sheet 1 of 1: Delete in its entirety replace with included sheet.

PAGE 56

4.4.2.27, Sheet 1 of 1, major 113, under requirement paragraph: Add "3.7.4".

PAGE 57

4.5.1.1,a.: Delete "Armed and Locked" and substitute "Safe and Locked".

4.5.1.2 Delete in its entirety and substitute:

"4.5.1.2 Dynamic conditions.

a. Electrical assembly Armed and Locked.

b. Remove 31.0 +0.5VDC on Power/Destruct.

c. Measure time to occurrence of fire TM (3.7.4.1.1).

d. Measure time for L&R fire circuits to function (3.7.4)."

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.26	Electrical Assembly Electrical Operation	OF 1	12630828
			NEXT HIGHER ASSEMBLY 12630801

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Fire circuit charging (Power/Destruct)		100%	3.7.2	Test
102	Fire upon arming after loss of power		100%	3.7.4.1/ 3.7.4	4.5.2/Test

NOTES:

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.

4.5.2.2,c.: Delete sentence and substitute: "Measure time for the fire circuits to function (3.7.4.1.2)."

Custodian:
Army - AR

Preparing activity:
Army - AR

(Project 1336-A577)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:36Z
Check the source to verify that this is the current version before use.